

CHANDRAYAAN 3

TIRANGA FLIES ON THE MOON



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Ten-year-old Veer is a very intelligent and curious boy. He and his Dadaji are great friends. They sleep in the same room. At bedtime, before falling asleep, Dadaji makes up interesting stories on any topic that he feels little Veer should be aware of. Veer loves to hear these stories. Veer's school was soon to hold a workshop on Space.



Veer wanted to actively participate in this workshop. He was very interested in learning about India's Space Research Programme. The topic constantly played on his mind. One night, after completing his homework, packing his bag for school, and getting his uniform ready, Veer jumped into bed and snuggled against Dadaji. Veer - Dadaji, tell me a story about space, the moon, and astronauts.

Dadaji's eyes twinkle. He had been expecting this question from Veer for quite some time now. He knew about the forthcoming workshop in the boy's school, and had seen his excited preparations.

Dadaji – What is so special about the moon today?

Veer – Dadaji, this morning while cleaning my room, I saw the picture of a spacecraft on the moon on the front page of a newspaper. There were also pictures of people celebrating. I want to know why this news was so important that day.





Dadaji - Once upon a time, a young princess wanted to go to the moon. The king didn't know how to fulfil his daughter's foolish wish. He promised a huge reward for anyone who would please his daughter. Many courtiers came forward with various ideas but none could please the princess.

One day, a saint came and took the challenge. He asked the king to bring his daughter before him. Saint- Tell me, little one, how big is the moon?

Dadaji - Veer, it was a very great day in the history of space research, as well as of India. Modi ji has rightly named 23 August as National Space Day. But first let me tell you a small story about the moon which I heard from my grandmother when I was your age.

Veer - Wow! Tell me quickly, Dadaji.

Saint - Tell me, little one, how big is the moon?

Princess - It is as wide as my index finger. When I lift it towards the moon, it completely covers it, and then I can't see the moon at all.



Saint - Do you know how high it is?

Princess - It's visible just above that tree in the palace garden. I think it is just as high as that tree.

Saint - What does the moon look like?

Princess - It is white and bright like silver.

Saint - How far away from you is it?

Princess - (softly singing) Chanda mama door ke... Oh!!!
The moon is far, far away.

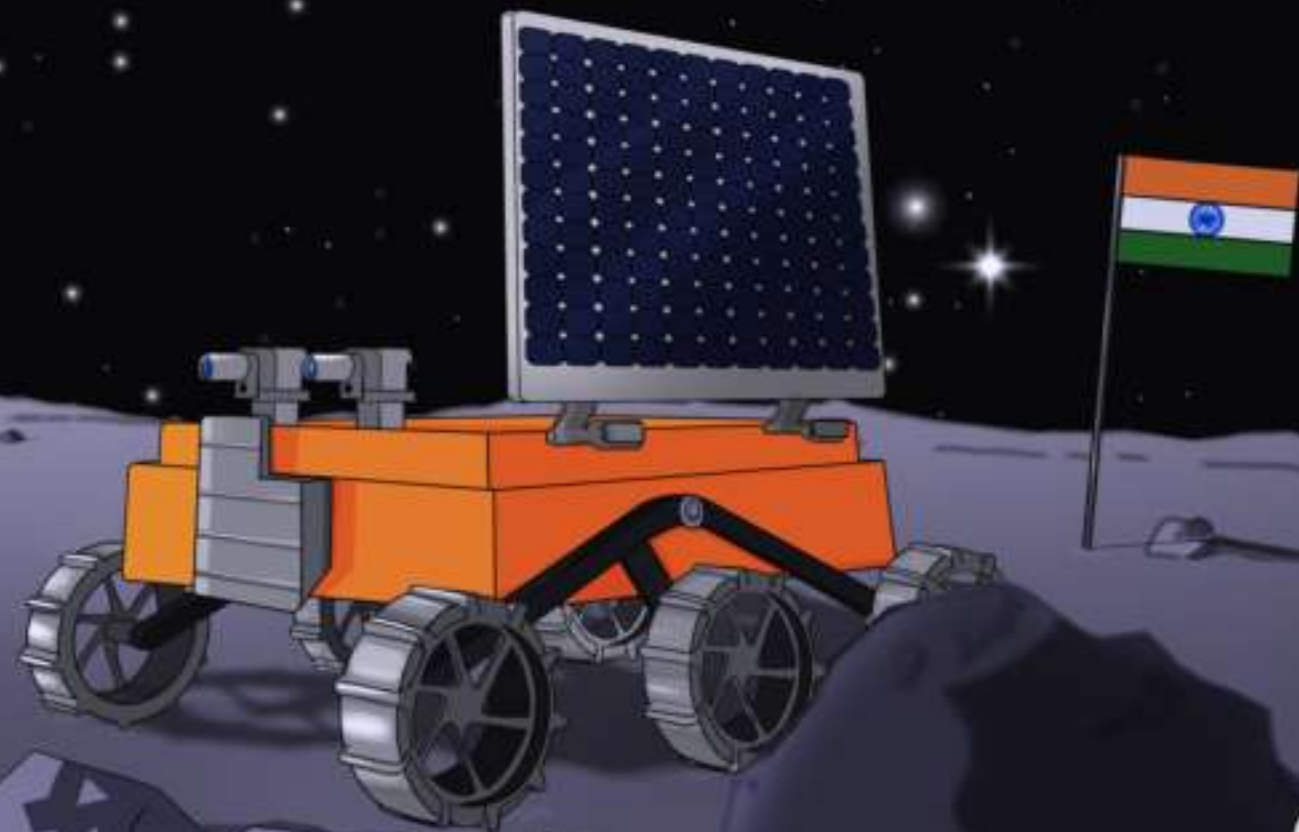
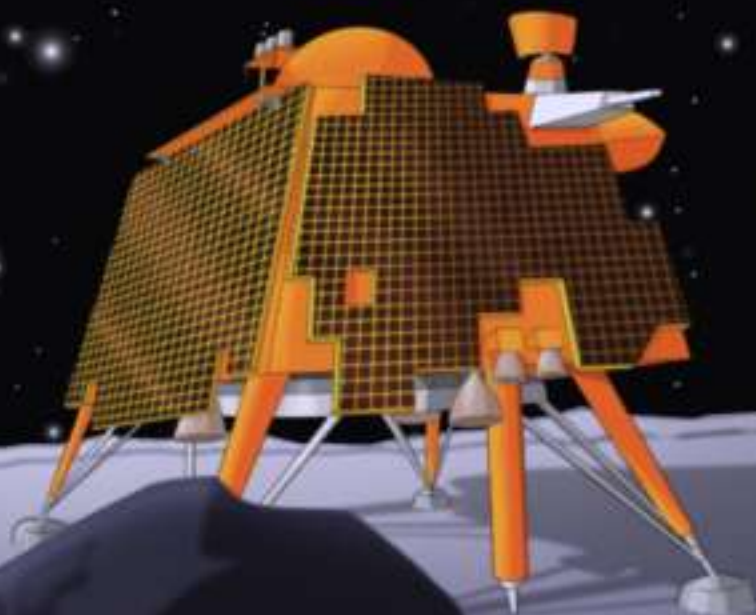
The king graciously awarded the saint for cheering up his daughter.

Veer was getting impatient. This was not the story he wanted to listen tonight.

Veer - What a silly princess. Who would believe this story?

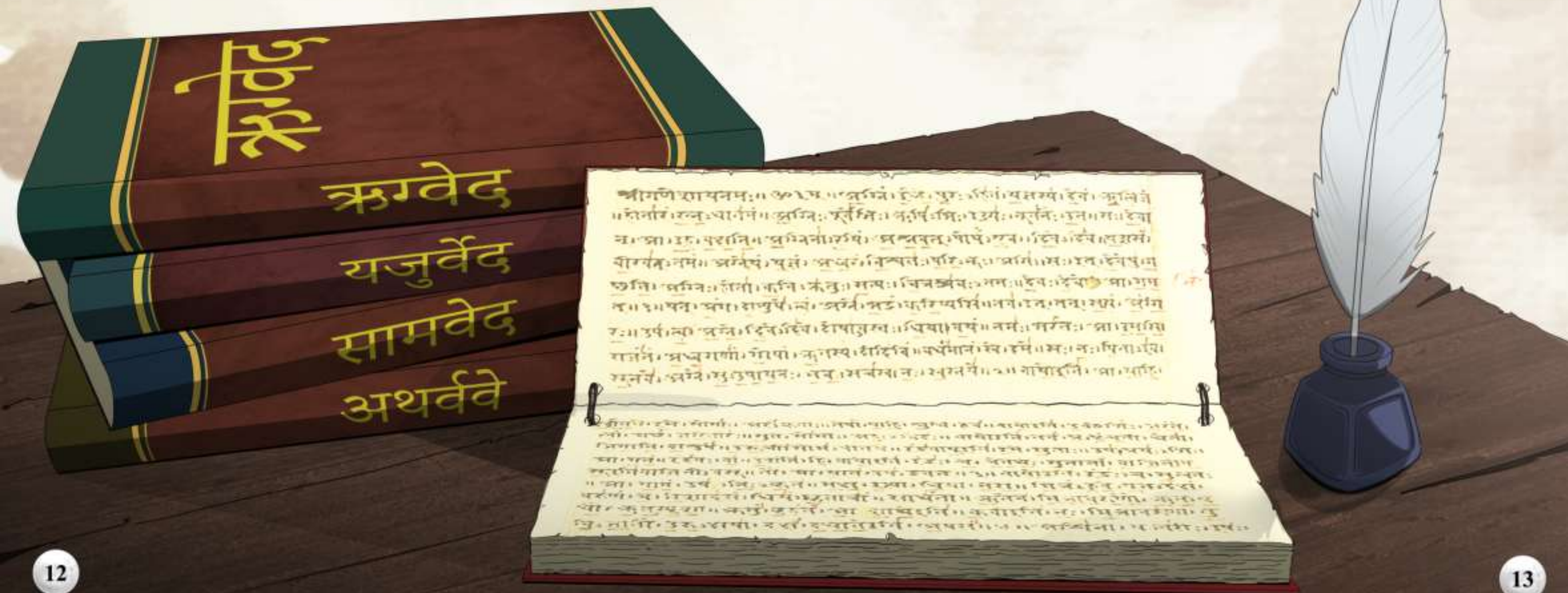
Dadaji - (Laughing) Veer, this is a story from the times when most folks knew nothing about the moon. You laughed at the princess because you know that many silly myths about the moon have been cleared. Now I will tell you all that you want to know about that headline.

Veer - Wow! I can't wait to know more. Tell me all, Dadaji. All the news channels too were very proudly broadcasting India's moonlanding, and everyone was so happy. Why was it such a big news? After all, hasn't India learnt about space from other countries who have already been to space?



Dadaji - Veer, this is what everyone has been made to believe. The fact is, that other nations have explored space just a few decades back. But many of our ancient Indian texts, like the Vedas, discussed about space and astronomy thousands of years ago. Do you know that India was once the vishvaguru, or the teacher of the world?

Veer's eyes widened. He had not known this about his country. He became even more alert and interested in what Dadaji was saying. The clever grandpa knew that Veer was keen to learn of the glory of India's past. He continued on the same lines.





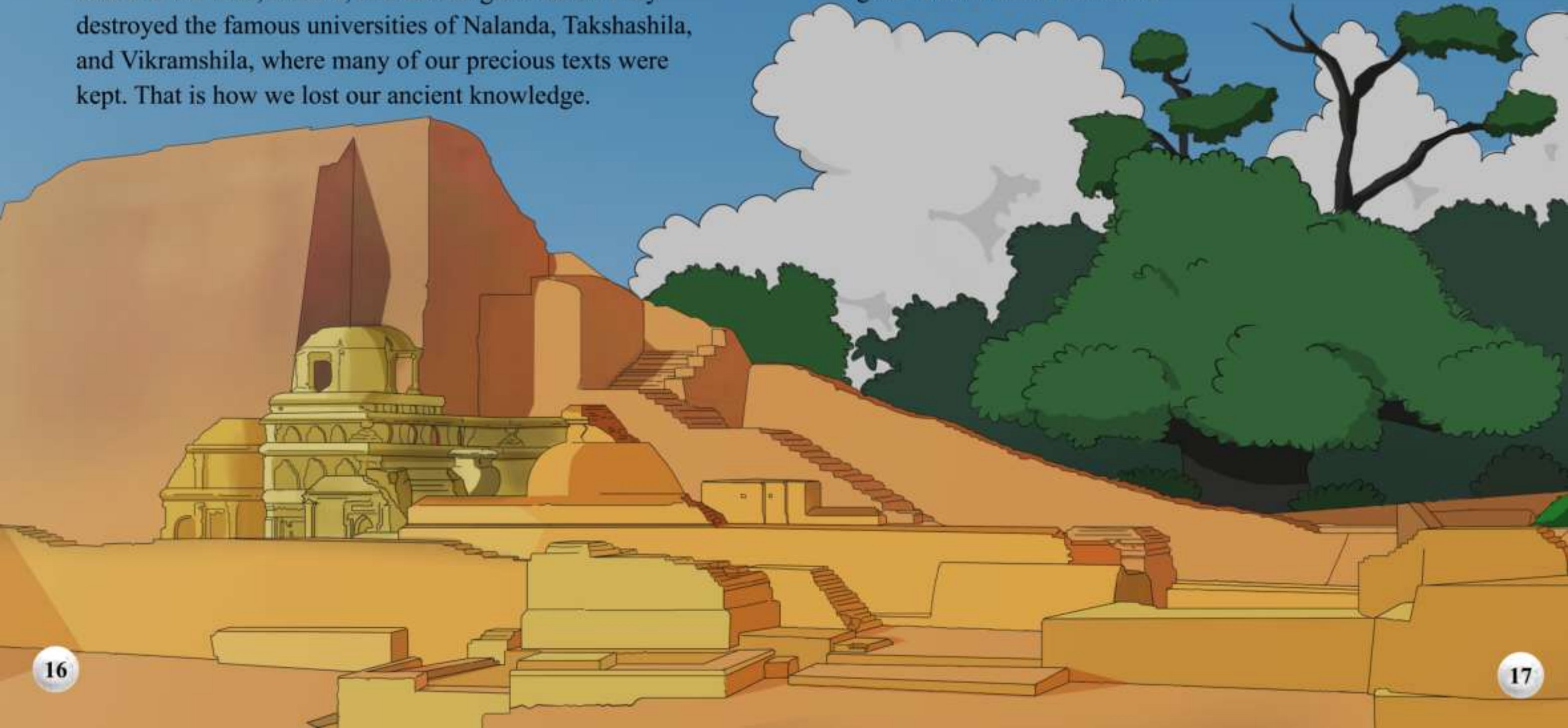
Dadaji - Thousands of years before people could not even dream of space and aircrafts, ancient Indian wise men had compiled the Vimaanshastra, or the science of flying. Gods and kings often covered long distances on their flying chariots.

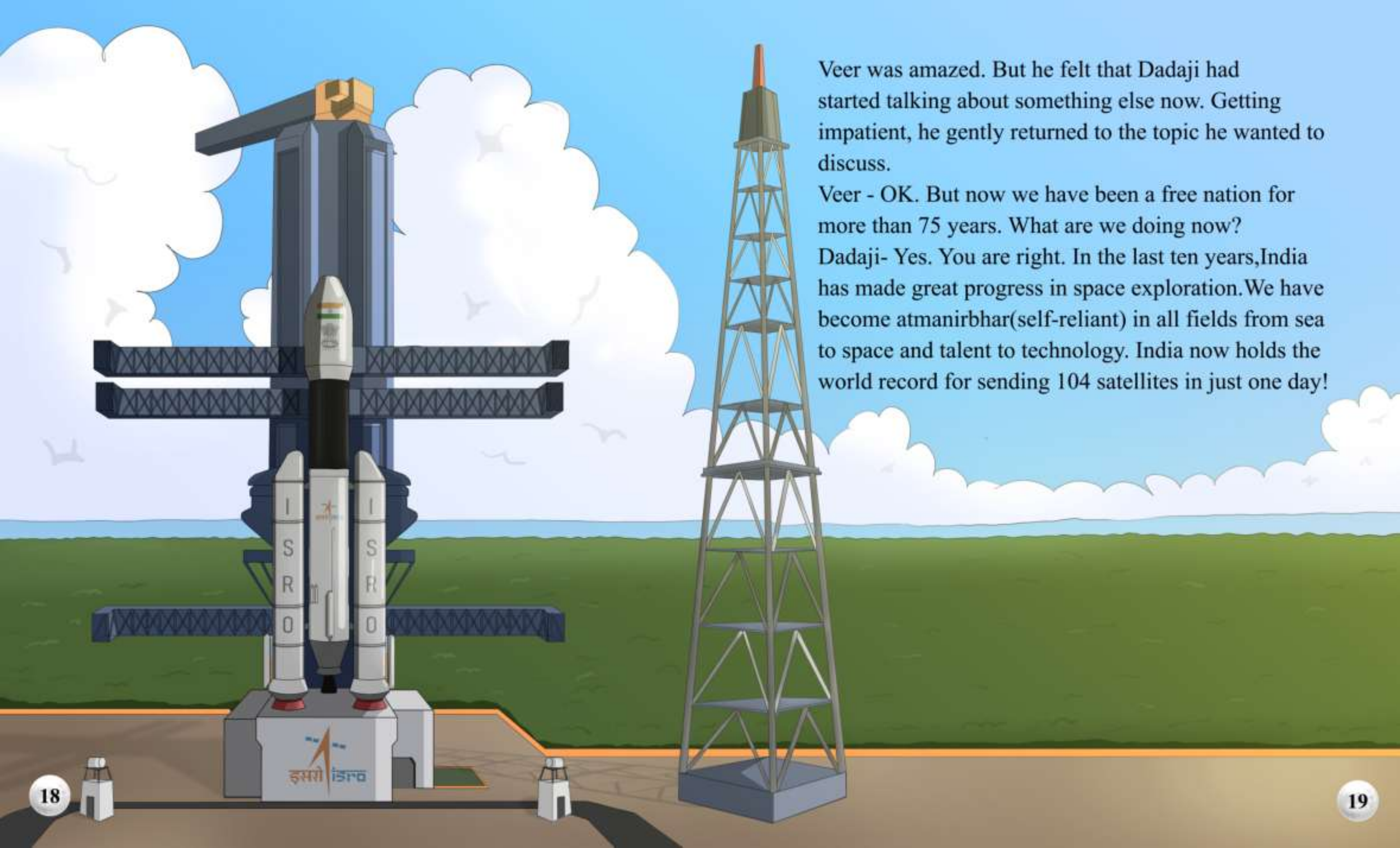
Veer - Yes, Dadaji. I remember! In the Ramayana too, Ravana kidnapped Sita and flew in his Pushpak Vimaan. But then, why did it take us so long to explore space?

Dadaji was expecting this question. He knew that Veer wanted to get into the root of the matter. He was prepared to answer.

Dadaji - For over fourteen hundred years, cruel invaders attacked our land, ruled it, and caused great harm. They destroyed the famous universities of Nalanda, Takshashila, and Vikramshila, where many of our precious texts were kept. That is how we lost our ancient knowledge.

Just imagine! Veer, the great mathematician Aryabhata who discovered 'zero' was an esteemed teacher at Nalanda University. The Chinese scholar Hiuen Tsang studied at Nalanda University and even became a teacher there, so great was its name and fame.





Veer was amazed. But he felt that Dadaji had started talking about something else now. Getting impatient, he gently returned to the topic he wanted to discuss.

Veer - OK. But now we have been a free nation for more than 75 years. What are we doing now?

Dadaji- Yes. You are right. In the last ten years, India has made great progress in space exploration. We have become atmanirbhar(self-reliant) in all fields from sea to space and talent to technology. India now holds the world record for sending 104 satellites in just one day!



Veer- I know. I saw the satellites video on youtube. It was so interesting. Tell me Dadaji, what is so special about Chandrayaan 3?

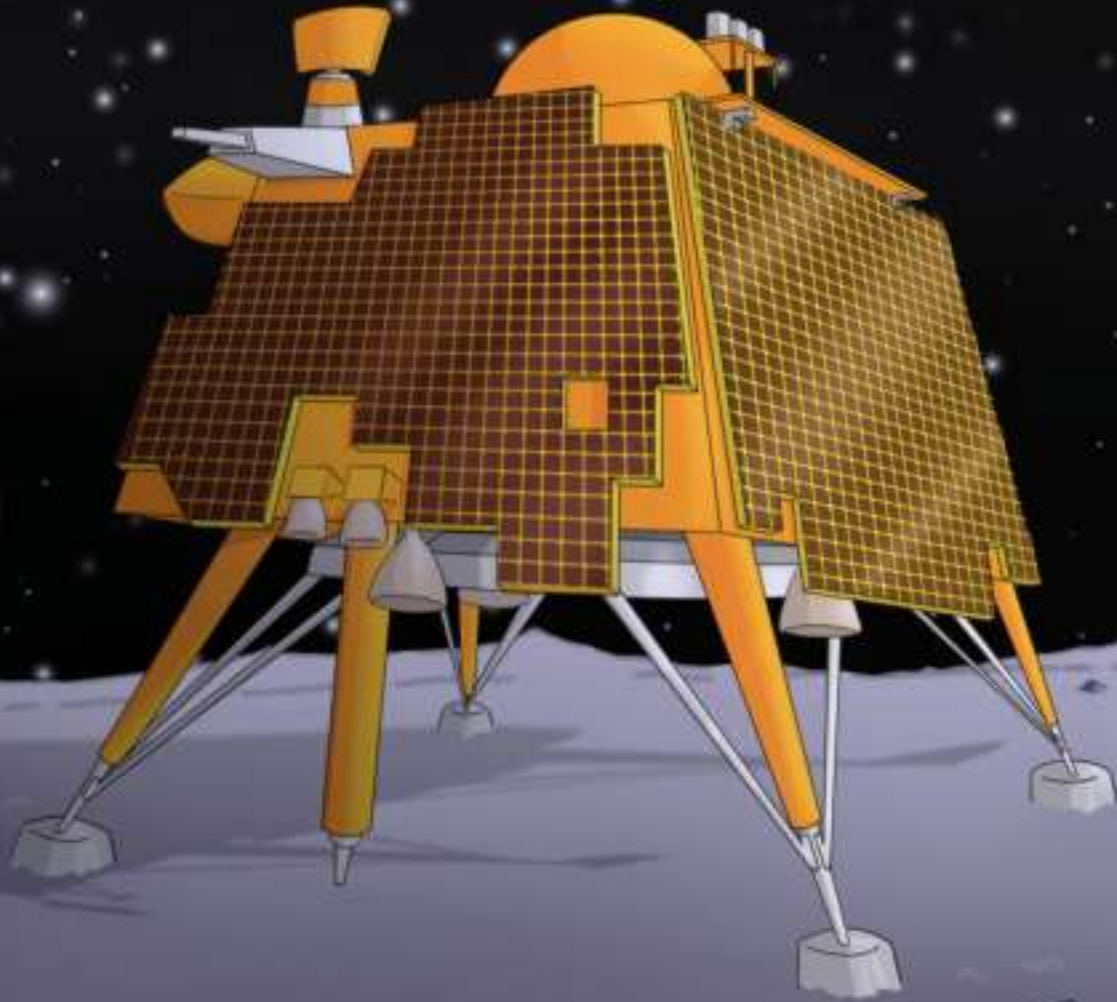
Dadaji - You have asked a very good question. I will explain in detail. Just then, Veer's mother entered the room with a tray with two glasses of hot milk for both of them. Dadaji took a sip.

Veer adjusted his pillow to be even more comfortable. The milk was too hot for him. He would wait a bit to drink it.

Dadaji - The landing of Chandrayaan 3 is a matter of great pride for us because India is the first country to successfully land on the south pole of the moon. Our scientists have achieved it at a very low cost of 620 million dollars which is almost half the cost of a Hollywood movie.

Veer's mind wandered to something he suddenly remembered as Dadaji spoke. He interrupted him.

Veer - Dadaji, tell me who are 'Vikram' and 'Pragyan'?
I heard something about them on T.V. Are they astronauts?



Dadaji - 'Vikram' is the name of the lander that has landed on the moon. 'Pragyan' is the name of the rover that will explore and try to find answers to unexplored facts about the moon. The lander has been named 'Vikram' in honour of our great scientist Dr. Vikram Sarabhai. He was the founding father of India's space programme.

Veer - I will surely read this book which tells about Vikram Sarabhai.



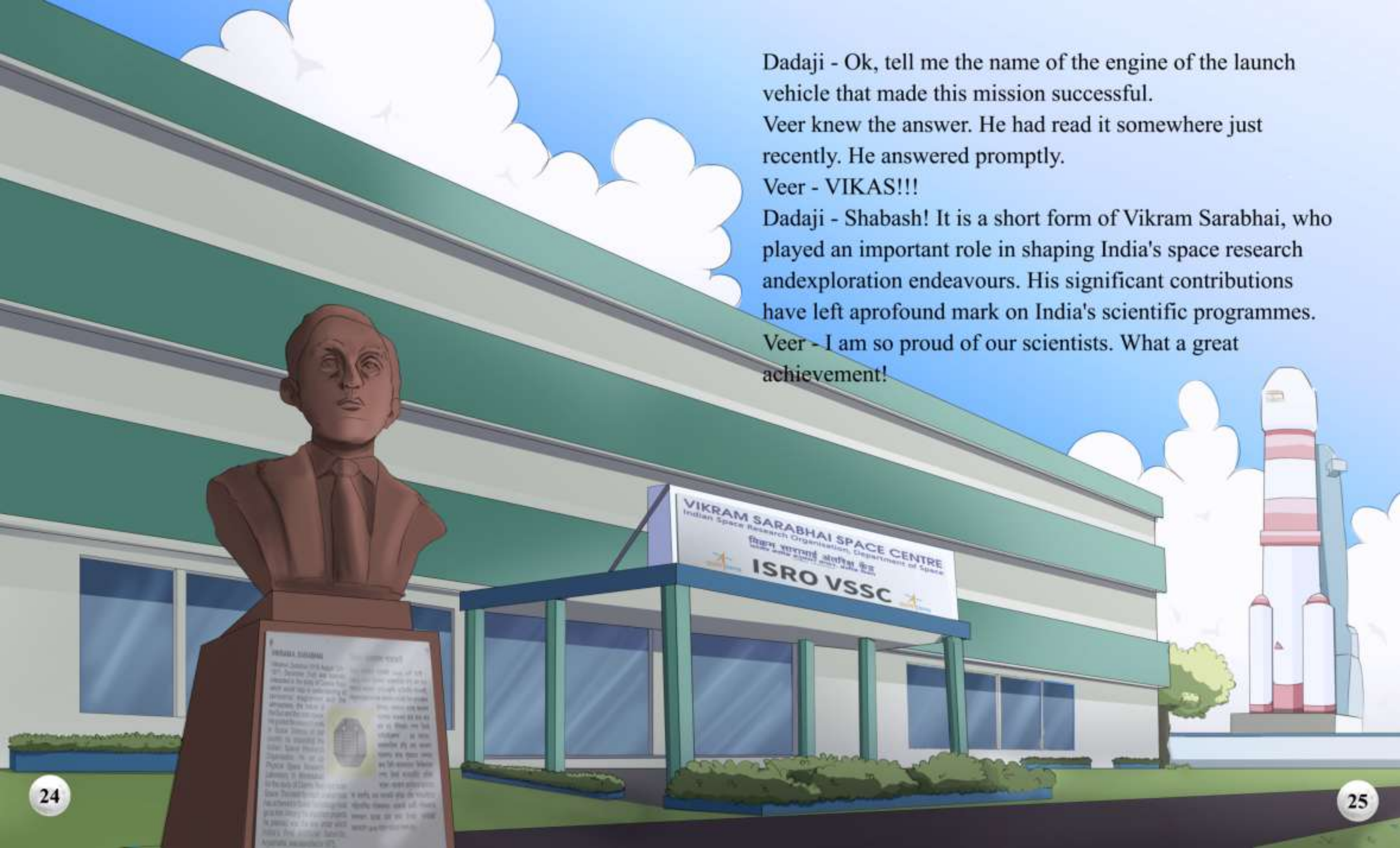
Dadaji - Ok, tell me the name of the engine of the launch vehicle that made this mission successful.

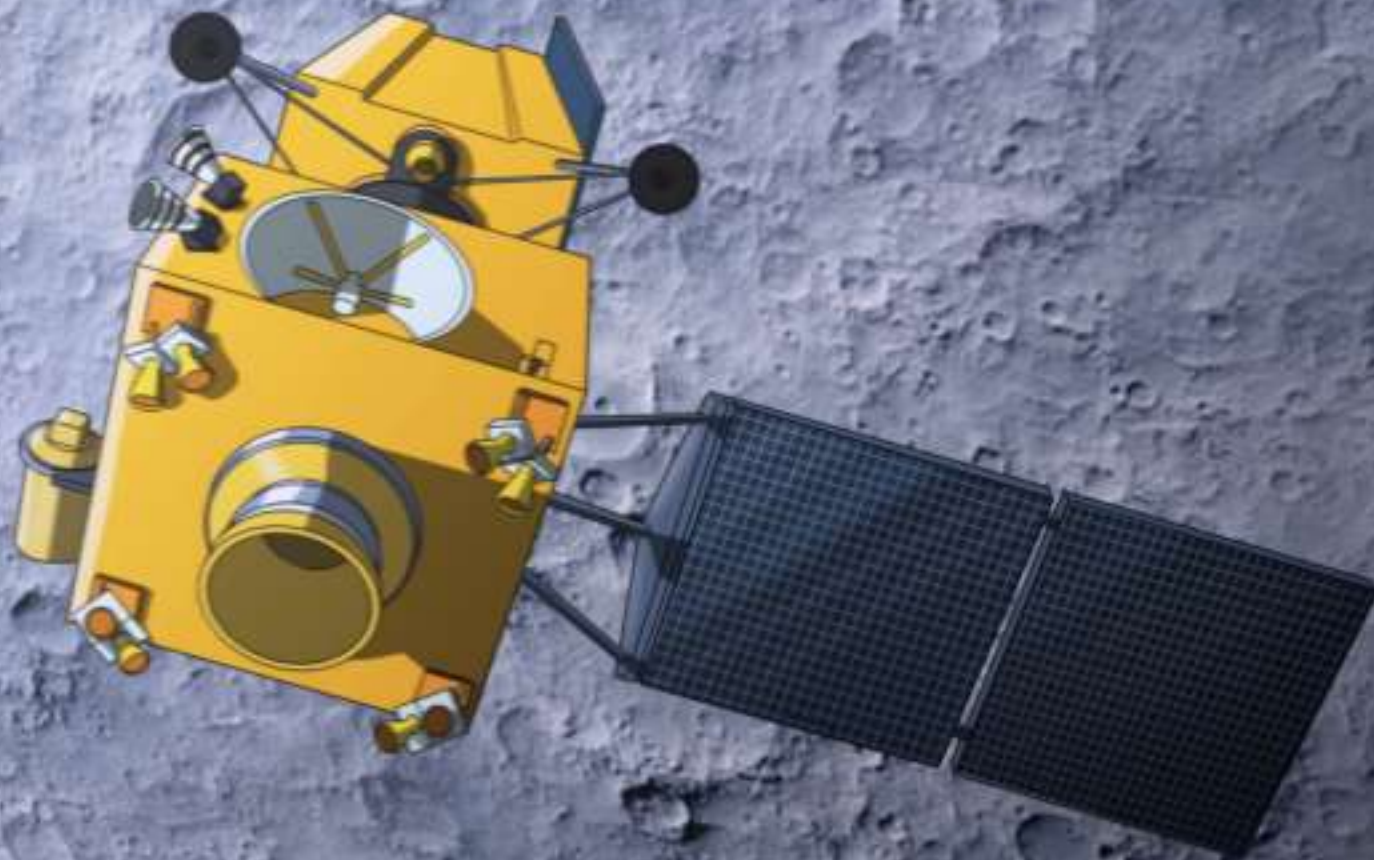
Veer knew the answer. He had read it somewhere just recently. He answered promptly.

Veer - VIKAS!!!

Dadaji - Shabash! It is a short form of Vikram Sarabhai, who played an important role in shaping India's space research and exploration endeavours. His significant contributions have left a profound mark on India's scientific programmes.

Veer - I am so proud of our scientists. What a great achievement!





Dadaji - Veer, first of all you should know that Chandrayaan is the name of the Indian Lunar Exploration Program.

India launched Chandrayaan 2 in 2019. The lander was on the right path, but unfortunately due to a small error, communication was lost in the final 300 meters, and it could not make a smooth landing on the surface of the moon. It crashed.



Veer nodded as he gulped down the now cold milk. He faintly remembered watching this news five years ago. Dadaji was surprised. How could Veer remember the event? He was just five back then.

Dadaji - Do you remember watching that landing of Chandrayaan 2? Our Prime Minister Shri Narendra Modi was also present in the operation control room to witness that historic moment. We were all so touched when Modi ji hugged and consoled the ISRO Chairman K Sivan and encouraged him to try again.

Veer - I can imagine how disappointing it must have been for the whole team.



Dadaji - No Veer. Our scientists did not sit and sulk at the setback. That same day they decided to correct the errors and planned Chandrayaan 3. They learnt from their mistakes and made a failure-proof model for the next mission and within four years succeeded in their mission.

We are proud of our scientists and of our leadership in showing faith in the scientific community. Over 100 women scientists were also involved in the mission and were in the control room at the time of the launch and during the landing on August 23.

Prime Minister Narendra Modi met and congratulated each one of them.

Veer - What will this rover do on the moon? Nobody stays there. How will it help us?

Dadaji - Ha! Ha! Ha! You are wrong, little boy. Like Chandrayaan 1 gave valuable inputs about the possibility of water on the moon, this mission will explore the properties of the moon's surface, its soil, gases, metals and minerals.

Veer smiled sheepishly. He wanted to show Dadaji that he was not a little boy. He knew a lot!

Veer - Dadaji, do you know the names of the places where Chandrayaan 2 and 3 landed on the moon?

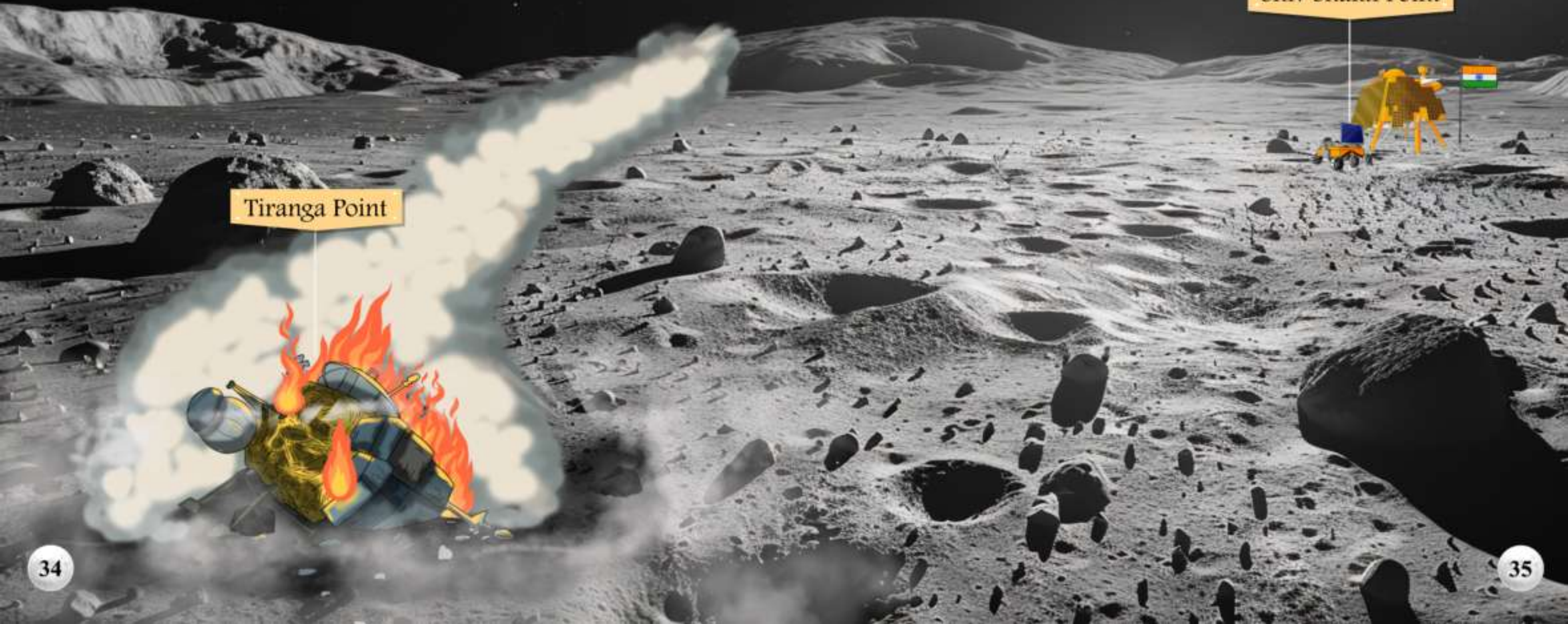


Dadaji understood that Veer had asked this question to impress him about his knowledge. So he played along.

Dadaji - You tell me. I don't remember exactly.

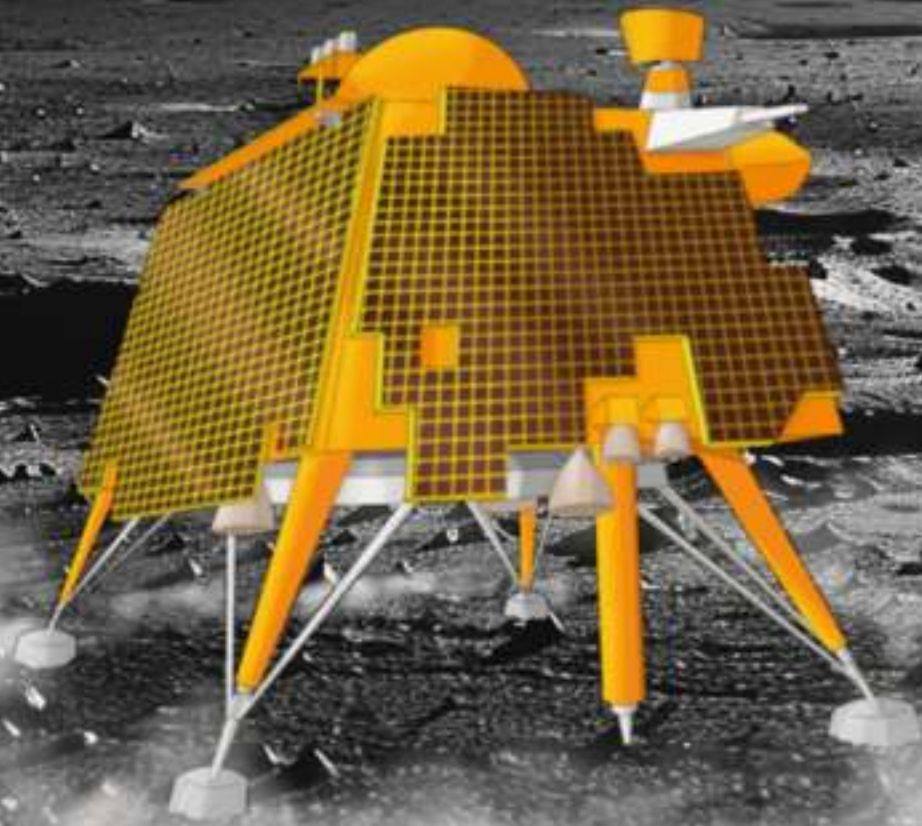
Veer - It is known as the 'Tiranga Point' where Chandrayaan 2 crash landed, and 'Shiv Shakti Point' where Chandrayaan 3 successfully landed.

Dadaji - Brilliant! Veer. Everything which our country does has a scientific and spiritual dimension to it. Shri Modi has named these two points himself after the success of Chandrayaan 3.



Dadaji - Veer, Shri Modi said, "The spot on the lunar surface where the Chandrayaan-2 left its footprints in 2019 will be known as 'Tiranga Point'. This will be an inspiration for every effort made by India. It will remind us that any failure is not final."

Veer - He is great. Dadaji, I remember Modi ji proudly holding the Indian flag and waving it on TV when Chandrayaan 3 landed successfully.



Dadaji - Yes, at that time Modi ji was in South Africa to attend the 15th BRICS summit but was keeping a close watch on the progress of the Chandrayaan 3. He is always there to support our space team.

Veer listened in wonder as Dadaji continued.



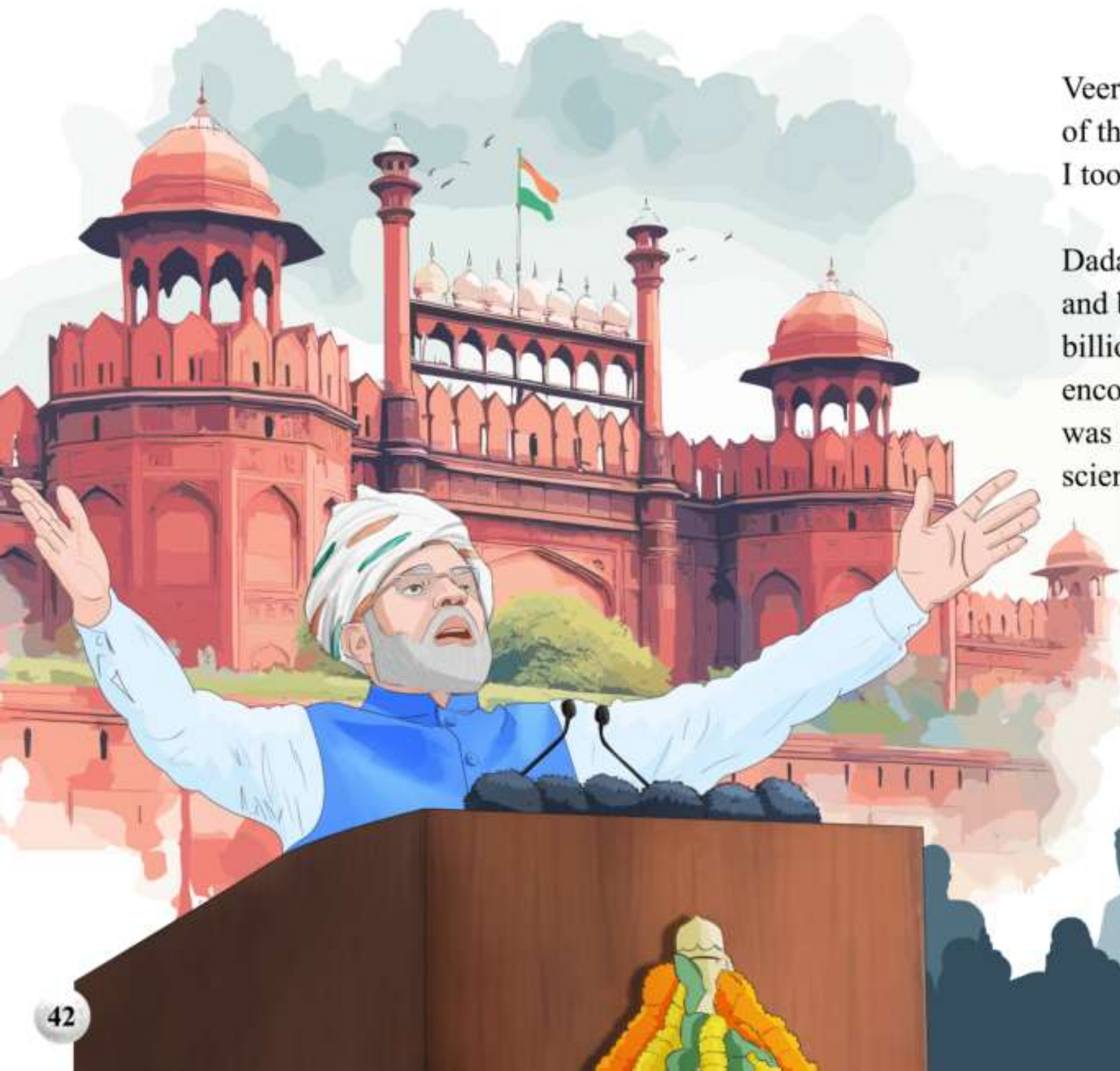
Dadaji - After his return, he himself went to the space centre in Bangalore and congratulated all the scientists personally. This kind of motivation boosts the morale of our young scientists and inspires them to achieve new heights.





Veer - Last Diwali I saw him on TV standing and talking with our soldiers at some snowy place.

Dadaji - Yes. That's his greatness. He never misses a chance to be with the Armed Forces and celebrate Diwali with them. He always shows utmost faith in their capabilities and boosts their morale with his presence.



Veer - Modi ji is my role model. His speech after the success of the Chandrayaan 3 mission was very impressive. Some day, I too want to become like him.

Dadaji - Yes, Veer. Modi ji has taken India to new heights and become a global leader. He is the Prime Minister of 1.45 billion people and carries their dreams and hopes. He encourages the youth of today. The 'Jai Anusandhan' slogan was first raised by PM Modi in January 2019 while imploring scientists to encourage research among students.



वसुधैव कुटुम्बकम्

ONE EARTH • ONE FAMILY • ONE FUTURE

Dadaji - I also loved his speech when he said –
“Chandrayaan Maha Abhiyan is a success not only for India but for the entire humanity.” It shows the world that India truly believes in Vasudhaiva Kutumbakam-meaning One Earth, One Family, One Future. That was the theme of India’s G20 Presidency held in New Delhi in September 23.



Veer - Are we going to launch another mission?

Dadaji - Certainly Veer. India is going to launch 'Gaganyaan' in 2024-2025 which is an Indian crewed orbital spacecraft that will take our astronauts to the space. It is going to be a huge step in the field of space exploration.

India's Space Vision 2047



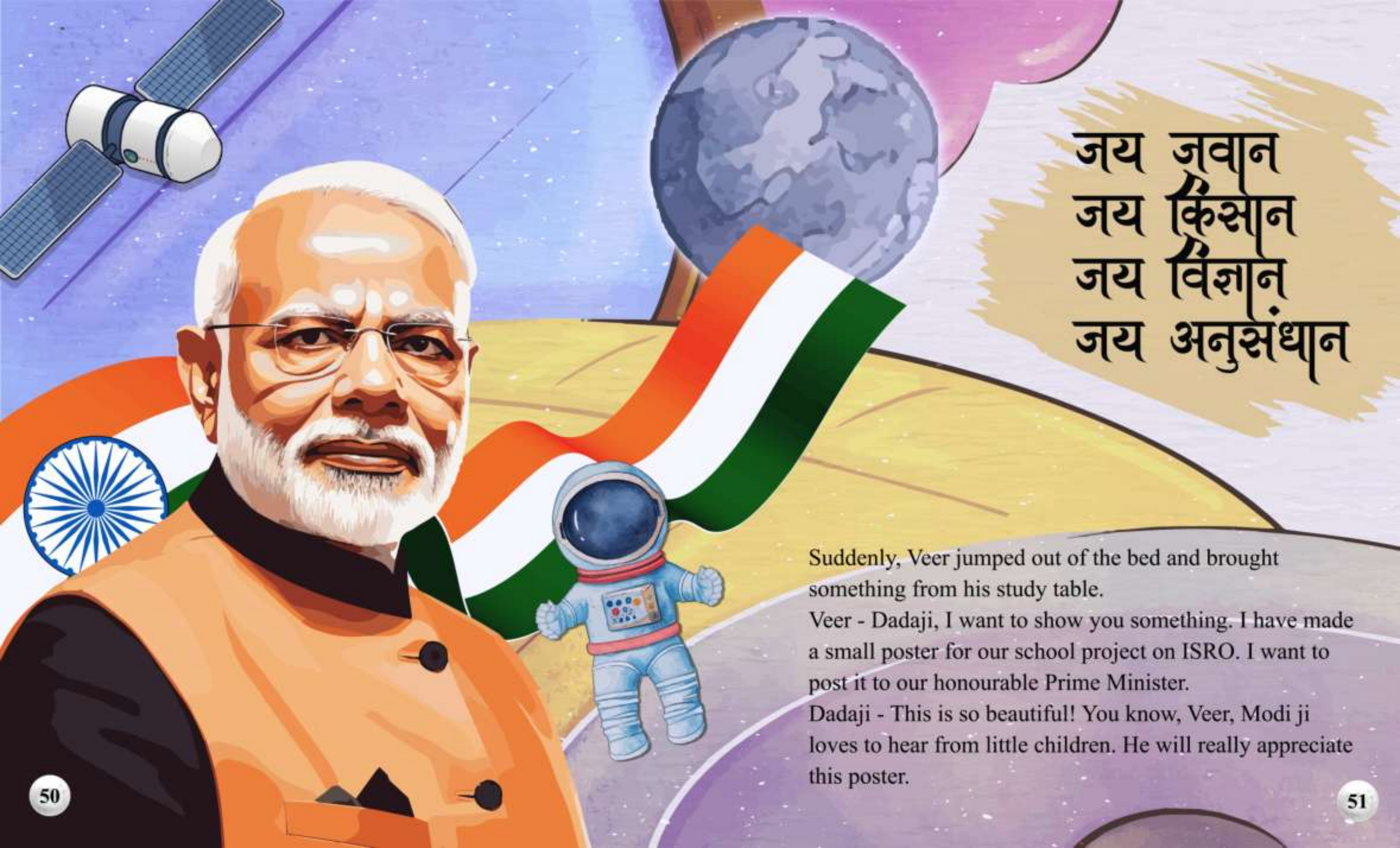


Veer - Dadaji, I also want to go to a space centre to see the spacecraft and learn about India's space programme.

Dadaji - Good idea. In your next holidays, we will go to the space centre at Sriharikota and the ISRO Headquarters at Bengaluru. We can also go to Hyderabad to see the Space Science Museum.

Veer - Dadaji, what is the full form of ISRO?

Dadaji - ISRO stands for Indian Space Research Organisation. It works to reap the benefits of outer space for India and mankind.



जय जवान
जय किसान
जय विज्ञान
जय अनुसंधान

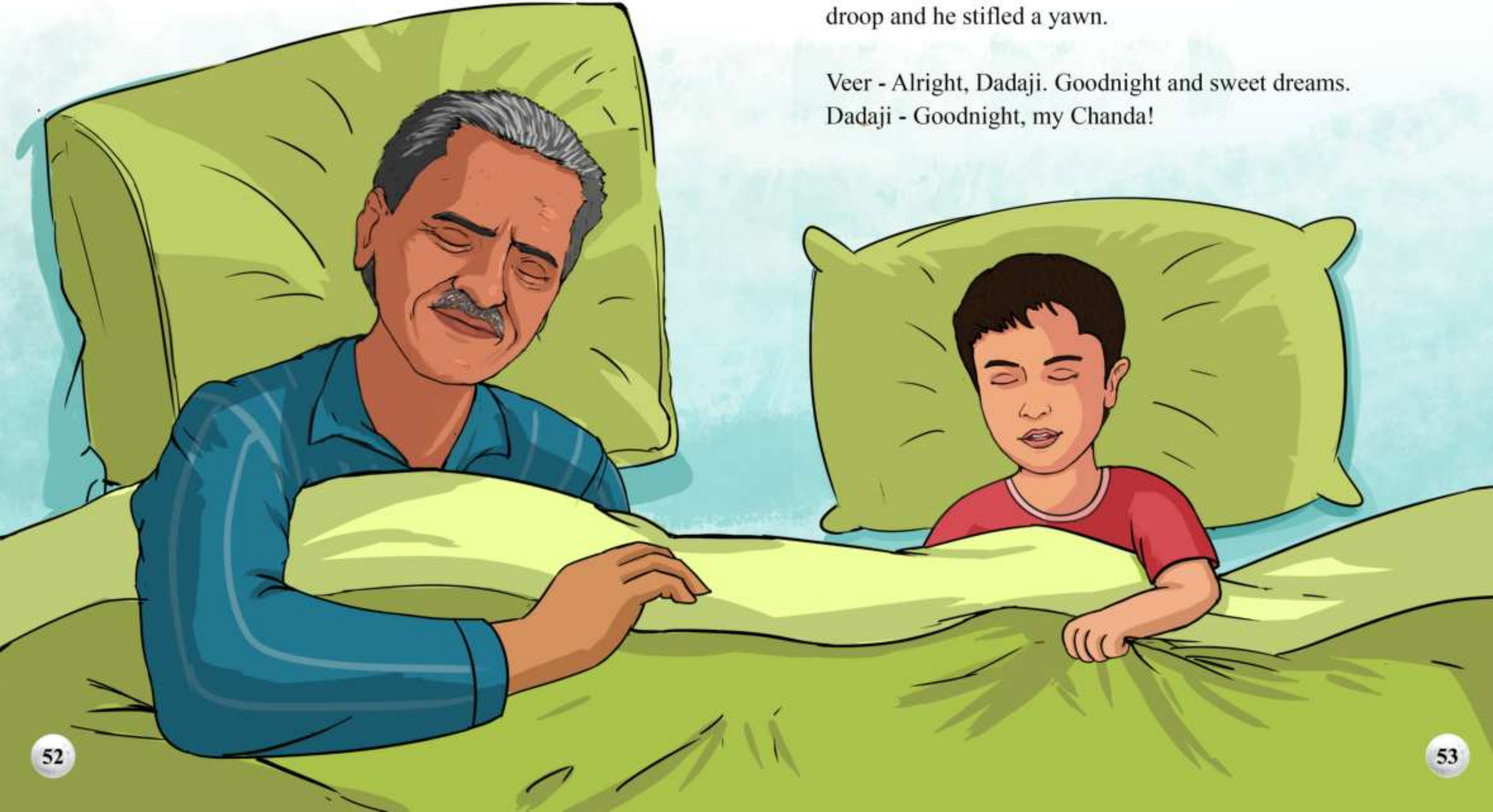
Suddenly, Veer jumped out of the bed and brought something from his study table.

Veer - Dadaji, I want to show you something. I have made a small poster for our school project on ISRO. I want to post it to our honourable Prime Minister.

Dadaji - This is so beautiful! You know, Veer, Modi ji loves to hear from little children. He will really appreciate this poster.

Veer was feeling sleepy now. His eyelids had begun to droop and he stifled a yawn.

Veer - Alright, Dadaji. Goodnight and sweet dreams.
Dadaji - Goodnight, my Chanda!



Excerpts from Modi's Speeches

1. To scientifically prove the astronomical principles in India's scriptures, the new generation should come forward with fresh perspectives for their study. This is essential for our heritage and also for science.
2. On 23 August when India unfurled the tricolour on the moon, that day will now be celebrated as National Space Day.
3. Today, from trade to technology, India is counted among the front-runners, or the 'first row', of nations. In the journey from the 'third row' to the 'first row', institutions like ISRO have played a significant role.
4. This tricolour point will become the inspiration for every endeavour of India. This tricolour point will teach us that no failure is final, success comes with strong determination.

5. In Chandrayaan-3, our women scientists have showcased the significant role of the power of women in our country. The 'Shivashakti' point on the moon will bear witness to India's scientific and philosophical contemplation for centuries to come.
6. This is the India of today. Fearless and resilient. It thinks anew, it thinks differently. Even in the darkest corners, it spreads rays of light across the world.
7. India is on the moon! We have our national pride planted there! We reached where no one had reached before. We did what no one had ever done.
8. In India, we all refer to the Earth as 'Mother' and the Moon as 'Uncle'. It was once said that Uncle Moon is very door (far away). Now, there will come a day when children will say, 'Uncle Moon' is just a 'tour' away.





9. The place where Chandrayaan-3 moon lander touched down will now be known as the 'Shivashakti' point. The resolve for the welfare of humanity is embedded in Lord Shiva, and 'Shakti' empowers us to fulfill those resolves.

10. India's Chandrayaan mission is also a vibrant example of women empowerment. Numerous female scientists and engineers have been directly involved in this entire mission. They have shouldered many critical responsibilities such as project directors, project managers, and many more, across different systems.

11. On 23 August, India and India's Chandrayaan have proven that some of the suns of determination also rise on the moon. The Chandrayaan mission has become a symbol of the courage of the new India, which strives to win at all costs, and indeed knows how to win in every circumstance.

12. When we witness such historic moments, we feel immense pride. This is the dawn of a new India. We made a commitment on Earth and realised it on the Moon... India is now on the Moon.

13. This day is a symbol of how victory is achieved by learning from defeat.

14. This moment signifies walking on the path of victory. This moment is the power of 1.4 billion heartbeats. This moment is the belief in new energy in India. This moment is the call of India's rising destiny.

15. After today, the myths associated with the moon will change, the narratives will change too, and the adages for the new generation will also change.

16. The successful Chandrayaan mission is not India's alone.

13. What are the three best lessons you have learnt from Modi ji's speeches?
14. Name three ancient world famous universities of India.
15. How many satellites did India send in a single day?
16. What do you know about Gaganyaan?
17. What are the leadership qualities you have learnt from our honourable prime Minister?
18. Where is the Space Science Museum located?

19. What is the name of the launch engine named after Vikram Sarabhai?
20. What did our scientists do after the crash of Chandrayaan 2?
21. What happened to India's ancient texts that were full of wisdom and knowledge?
22. Which famous scholar taught at Nalanda University?
23. Where was PM Modi on the day when Chandrayaan 3 landed successfully on the moon?
24. What valuable inputs did Chandrayaan 1 give?



For Answers, Refer

- | | |
|-------------------------|---|
| 1. (Pg 8-9) | 13. (Pg 56, 57, 58, 59) |
| 2. (Pg 50-51) | 14. (Pg 18-19) |
| 3. (Pg 24-25) | 15. (Pg 20-21) |
| 4. (Pg 36-37) | 16. (Pg 48-49) |
| 5. (Pg 36-37) | 17. (Go through every page of the book) |
| 6. (Pg 16-17) | 18. (Pg 50-51) |
| 7. (Pg 22-23) | 19. (Pg 26-27) |
| 8. (Pg 24-25) | 20. (Pg 32-33) |
| 9. (Pg 46-47) | 21. (Pg 18-19) |
| 10. (Pg 48-49) | 22. (Pg 18-19) |
| 11. (Pg 44-45) | 23. (Pg 40-41) |
| 12. (Pg 56, 57, 58, 59) | 24. (Pg 34-35) |

About the Author

Yuvraj Malik is presently the Director, National Book Trust, India (Ministry of Education, Government of India) since 2020. Before joining NBT India, he has served with Ministry of Defence, Ministry of Home Affairs, Raj Bhawan, Jammu & Kashmir, United Nations Missions in Africa.



Yuvraj Malik

He is also the Fair director / CEO of New Delhi World Book fair which is the biggest B2C book fair in the world. He has also been Chairman of Commission for Scientific & Technical Terminology (CSTT), Ministry of Education, Government of India.

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Panch Praan for a Prosperous India

The period of the next 25 years is the Amrit Kaal leading up to a century of independence. Let us make it a defining era in the history of our country. To do so, let us embrace the 'Panch Praan', five principles that will take India to greater heights.

Goal of Developed India - Viksit Bharat

Remove any trace of colonial mindset

Celebrate our heritage

Strengthen unity

Focus on duties



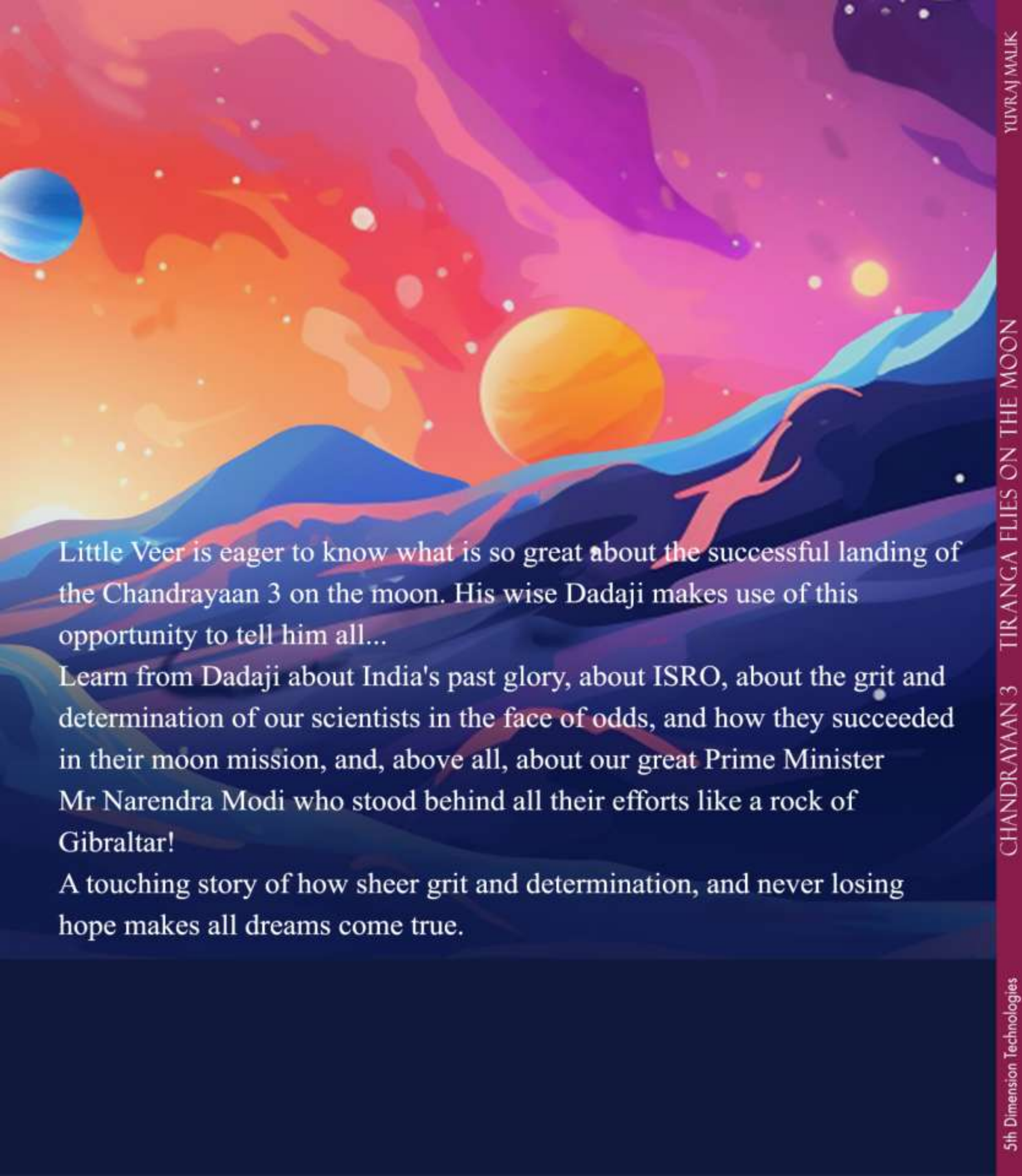
PART IVA

Article 51A

Fundamental duties

It shall be the duty of every citizen of India –:

- (a) to abide by the Constitution and respect its ideals and institutions, the National Flag and the National Anthem;
- (b) to cherish and follow the noble ideals which inspired our national struggle for freedom;
- (c) to uphold and protect the sovereignty, unity and integrity of India;
- (d) to defend the country and render national service when called upon to do so;
- (e) to promote harmony and the spirit of common brotherhood amongst all the people of India transcending religious, linguistic and regional or sectional diversities; to renounce practices derogatory to the dignity of women;
- (f) to value and preserve the rich heritage of our composite culture;
- (g) to protect and improve the natural environment including forests, lakes, rivers and wild life, and to have compassion for living creatures;
- (h) to develop the scientific temper, humanism and the spirit of inquiry and reform;
- (i) to safeguard public property and to abjure violence;
- (j) to strive towards excellence in all spheres of individual and collective activity so that the nation constantly rises to higher levels of endeavour and achievement.
- (k) to provide opportunities for education by the parent the guardian, to his child, or a ward between the age of 6-14 years as the case may be.



Little Veer is eager to know what is so great about the successful landing of the Chandrayaan 3 on the moon. His wise Dadaji makes use of this opportunity to tell him all...

Learn from Dadaji about India's past glory, about ISRO, about the grit and determination of our scientists in the face of odds, and how they succeeded in their moon mission, and, above all, about our great Prime Minister Mr Narendra Modi who stood behind all their efforts like a rock of Gibraltar!

A touching story of how sheer grit and determination, and never losing hope makes all dreams come true.